

# Adhithi Mudaliyar

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## EDUCATION

TAMPA, FL

### University of South Florida

Grad: May 2028

*Bachelor of Science in Computer Science & Economics*

GPA: 3.80/4.00

- **Relevant Coursework:** Natural Language Processing, Probability & Statistics, Data Structures & Algorithms, Intro to Machine Learning, Microeconomics

## TECHNICAL SKILLS

**Languages:** Python, SQL, Java, C, C++, JavaScript

**Machine Learning & Statistics:** scikit-learn, XGBoost, PyTorch, RDKit, Hugging Face Transformers, hypothesis testing, regression, classification, cross-validation, calibration, SHAP

**Data & Cloud:** Pandas, NumPy, PostgreSQL, Spark, Airflow, Databricks, AWS (S3, SageMaker), Docker

**Visualization & Workflow:** matplotlib, seaborn, Streamlit, Power BI, Git/GitHub, Azure DevOps, pytest, Microsoft Excel

## EXPERIENCE

### Research Assistant

Sept. 2026 – Present

*Dr. Shannon Bailey's Lab, USF Health CAMLS*

Tampa, FL

- Build **Python/Pandas pipelines** to standardize simulation errors, completion times, and device conditions for experimental analysis; document reproducible methods for interdisciplinary collaborators.
- Validate experimental logging and simulation behavior through **end-to-end testing** of an XR medical-training platform.

### Student Researcher

Apr. 2026 – Present

*University of South Florida, Prof. Yu Sun*

Tampa, FL

- Reconciled **ADNI and HABS-HD** schemas and variable definitions; built **Python/Pandas pipelines** with automated **data-validation checks** to clean, impute, normalize, and harmonize **32 multimodal features** across **5 research teams**.
- Developed **scikit-learn classification workflows** with reproducible **five-fold cross-validation**, feature importance, and error analysis; achieved **81.5% and 97.2% held-out accuracy** on ADNI and HABS-HD, respectively.

### Software/Firmware Engineering Intern

May 2026 – Aug. 2026

*Elevate Healthcare*

Sarasota, FL

- Built **Python/Linux tooling** and C/C++ firmware to integrate and test **4 healthcare simulation subsystems**.
- Worked in **Agile/SCRUM sprints**, participating in sprint planning and retrospectives; used **Git and Azure DevOps** to track fixes, validate device behavior, and document reproducible integration tests and handoff procedures.

## PROJECTS

### ToxiScope | Python, RDKit, PyTorch, scikit-learn, RAG, Streamlit

Jan. 2026

- Built **Tox21 assay-activity models** with **Morgan fingerprints** and **active learning**; matched random selection's ROC-AUC with **40% fewer labeled compounds** using **scaffold splits** and an untouched test set.
- **Fine-tuned BiomedBERT** with Hugging Face and built **PubMed RAG** with **BM25 + dense retrieval**; checked citations across **50 queries** and presented results through **Streamlit**.

### Diabetes Risk Intelligence System | XGBoost, SHAP, FastAPI, Docker, AWS SageMaker/S3, pytest, Streamlit

May 2025

- Achieved **0.89 held-out ROC-AUC** with ensemble models, class-imbalance handling, **probability calibration**, and tuning; deployed containerized **FastAPI inference** with **SHAP** and Streamlit, serving predictions in **under 120 ms**.
- Built **MLOps workflows** to automate training and evaluation, validate data with **pytest**, deploy through **SageMaker with S3 model artifacts**, and monitor model performance and latency.

### MedLens | Python, SQL, Airflow, Spark/Databricks, AWS S3, XGBoost, Hugging Face, Streamlit, Power BI

Sept. 2026

- Built **Airflow/S3 pipelines** that processed **5 million records across 3 annual releases**, integrating **CMS Part D** data and patient reviews; used **Spark/SQL in Databricks** for standardization and **data validation**.
- Analyzed spending growth and cost per prescription; evaluated **1-year-ahead XGBoost forecasts** on **time-based holdouts**, achieving **10% lower MAE** than a last-year-spending baseline.
- Built **Power BI dashboards** for regional medication use and spending trends, and an interactive **Streamlit** interface with **NLP summaries** of sourced patient-review themes.

## LEADERSHIP

### Technical Lead

Apr. 2025 – Present

*Association for Computing Machinery, USF Chapter*

Tampa, FL

- Lead a **10-member technical team** delivering AI and software workshops to **50+ students weekly**; mentor students in programming, debugging, testing, and **Git-based collaboration**.
- Directed integration of a gesture-controlled robotic arm and secured **\$520 in project funding**.